

CHILLING/FREEZING STRESS

Cause and Consequence in Maize Planting Dates in GermanyP. S. Parker^{1,2}, J. S. Shonkwiler³ & J. Aurbacher¹¹ Justus Liebig University, Giessen, Germany² Leibniz Zentrum for Agricultural Landscape Research, Müncheberg, Germany³ University of Georgia, Athens, GA, USA**Keywords**

adaptation; climate change; maize; panel regression; planting dates; risk

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Accepted July 7, 2016

doi:10.1111/jac.12182

Abstract

An important part of agricultural adaptation is the timing of crop sowing dates, affecting yields and the level of risk incurred during a particular season. Cold stress is especially relevant in maize, *Zea mays* L., so that the timing of planting in the spring is a tactical response to short-term weather, but is also subject to strategic planning with regard to longer-term climate. Both factors compare the potential implications of cold stress to the additional yield obtainable through earlier planting. New cultivars suited to growing conditions in Europe and generally increasing spring temperatures have enabled earlier planting, but it is still dependent on short-term weather during the planting period. In the context of field-level decision-making, a panel regression is used to estimate the relationship between weekly local temperature and precipitation and planting dates at specific sites throughout Germany. Next, localised weather data and planting behaviour are linked to yields at the district (*Landkreis*) level to show the effects of planting date on yield. Based on these relationships optimal planting dates are explored with some associated costs and benefits. Results show a trend towards earlier planting that follows observed increasing spring temperatures and the availability of more cold-tolerant cultivars but this advance is buffered by the increasing severity of minimum temperatures during a critical period. Earlier planting potentially increases yield but this is offset by additional management costs and risk. A robust and simple depiction of farmer behaviour in climatic, technological and economic context can help to understand trends in crop management and productivity that effect agricultural landscapes.

Introduction

Strategic crop management entails the direct engagement of farmers with their environment. One of the most pliable decisions in this regard is when to plant a field, which especially with regard to the increasingly economically important German crop *Zea mays* L., is a weighed compromise between expected yield and risk. Earlier planting, in favourable conditions, leads to earlier development and better synchronisation of maximum leaf area, hence radiation interception, with the seasonal maximum of incident radiation (Stone et al. 1999). It also affords a longer growing season in which biomass can accumulate and the crop can mature (Darby & Lauer 2002). Concurrent with these benefits is an increased risk of damaging cold weather that is

mitigated by both delayed planting, use of cold-hardy cultivars and other tactics (Farooq et al. 2008a,b,c). Earlier ripening cultivars, in the context of variable annual temperatures, mitigate late-season risk by increasing the odds of there being sufficient cumulative temperature for crop maturity before unfavourable fall weather. Planting later and using later-ripening cultivars increase the likelihood that the optimal growth period and, consequently, yield are not realised. An approach to reduce the risk of cold-weather stress is to delay planting and use fast-ripening cultivars, while the pursuit of maximum yield would mean planting late varieties as early as possible. Kucharik (2006) attributed the trend towards earlier maize planting more towards breeding and technological changes than to increasing spring temperatures. Kucharik (2008) also

determined April temperatures and precipitation to be primary drivers of maize planting dates in the Central U.S. Further, planting date and August temperature were both significantly and inversely related to yield. Earlier planting increasing yield was also found by Fairey (1983), Staggenborg et al. (1998), Oluwaranti et al. (2008) and Kolo et al. (2012).

Modelling spring planting dates, especially in the context of climate change, have been well examined (e.g. Leenhardt and Lemaire 2002, Kaukoranta and Hakala 2008, Sacks and Kucharik 2011, White et al. 2011, Olesen et al. 2012, Van Oort et al. 2012, Waha et al. 2012, Aurbacher et al. 2013, Eitzinger et al. 2013). Studies to simulate regional or time series crop yields, as with crop growth models, have often held planting dates static over space and/or time to focus on model sensitivity to other inputs (Leclère et al. 2013, Bassu et al. 2014, Mast et al. 2014, Nendel et al. 2014). To reduce the uncertainty that is accentuated by omitting planting date dynamics in biophysical modelling, results from a planting date model (Sacks et al. 2010) can be used as input in yield simulation studies (Deryng et al. 2011). A model that can accurately estimate specific location-year planting dates can help account for climate-responsive farm management, the lack of which is a source of uncertainty in agroecosystem models. The benefit of calibrated planting dates must outweigh the cost, so that a simple method such as that presented here, which is based only on weather input, is advantageous. For this case study of Germany, the economic impact of planting date variability was examined by relating the spatiotemporal dynamics of maize planting to respective district yields with a simple statistical model of how management and weather influence regional yields.

The phenology of German agricultural species has been thoroughly examined in relation to climatic records (Chmielewski et al. 2004, Menzel et al. 2006a, Estrella et al. 2007, Ma et al. 2012, Siebert and Ewert 2012, Bock et al. 2013). And while these studies have done much to explain the relationships between weather and phenology, they are not designed as predictive models and have not been related to crop yields. Menzel et al. (2006b) raised the question of why farmers in Germany appear slow to respond to a clear shift in climate towards warmer conditions throughout the year. To answer this question it may be necessary to more thoroughly examine the relationship between site-specific annual planting dates and the respective local daily weather, together with some consideration of farmer strategy, as in Maton et al. (2007).

This study uses panel analysis of twenty recent years of planting dates as determined by local weather to suggest what is affecting farmers in their annual management decision-making and how this combines with seasonal weather to affect yields. Analysis is based on three datasets, daily

weather conditions and annual crop phenology both from the German Weather Service (*Deutscher Wetterdienst*, DWD) and annual records of District (*Landkreis*) crop yields. Supplemental information including some background on breeding trends, further explains the existing agronomic context.

Geography

Germany ranges from about 47° to 55° N latitude and 6° to 11° E longitude, encompassing maritime lowlands in the North, Central uplands, the Rhein River basin in the Southwest and foothills and mountains in the South. Although a trend towards higher annual temperatures has been generally observed, a clear distinction is observable over the last decades wherein the southern part of the country has seen faster warming during the cold season while increasing mean temperatures in summer months have been more pronounced in northern areas, and further, rainfall during the same period has been generally increasing in the West and decreasing in the East (Hattermann et al. 2013). The country has been partitioned into 52 soil-climate zones (*Boden-Klima-Räume*, Roßberg et al. 2007) based on generalised homogeneity of soil properties and weather patterns. The zones in which observational data satisfying the selection criteria are available are included in the model as binary variables to capture geographic factors including soil, latitude, longitude and elevation.

Genetics

Although there is no consensus on the minimum temperature required for maize growth, Birch et al. (2003) concluded 8 °C, while Knappenberger and Knoller (2012) and Bonhomme (2000) maintain 10 °C. The latter explains that intraspecies variation appearing to contradict this can be attributed to photoperiod sensitivity, so that long-day genotypes are slowed in development during coincidentally lower temperatures. Subject to cultivar variation, generally, temperatures under 20 °C limit the ability of maize to reach its genotype-determined yield potential, temperatures under 5 °C result in injury that reduce yield potential and negative (Celsius) temperatures are lethal (Greaves 1996).

For several decades, European maize breeding has been crossing North American dent with European flint lines to produce new hybrids (Reif et al. 2005), so that German farmers have at their disposal a large variety of maize cultivars, among other traits differing in early vigour, (cold-resistance) and time to maturity, with which farmers can adjust cropping strategies according to local conditions. Increased early vigour facilitates earlier planting and thus an extended growing season at a given level of risk of cold-

weather stress. The standard FAO maturity rating of maize relates moisture content in the harvested crop to accrued temperature over time and is therefore an estimate of the time required between planting and maturity, a critical factor for farmer's planning a season (Amler 2006). Relatively faster-ripening cultivars permit later sowing while maintaining the expected date of maturity in the fall (Strigens et al. 2012). This is another way to mitigate the risk of early season cold stress. To meet the needs of Germany's maize farmers the Federal Plant Variety Office (*Bundessortenamt*, BSA) had 96 maize cultivars registered in 1992, increasing to 200 by 2011. Figure 1 shows the trends within registered cultivars over the last thirty years first towards earlier, then later ripening (lower FAO maturity rating) and greater early vigour, (less cold-sensitive) as evaluated by the BSA (2010). Maton et al. (2007) show how planting date and maize earliness selection are both strategic decisions made by farmers. Farmer demand has to some degree driven these trends in commercial cultivars. Their availability has enabled adaptive field practices that reverberate throughout German agricultural landscapes. Cultivar selection and its link to planting date decisions have ecological and economic implications that can be traced through integrated farm modelling (Troost and Berger 2014). Demonstrated

in the following is a method to model an important management decision and link it to yields in a simple model accounting for geographic heterogeneity throughout Germany.

Methods and Materials

The German Weather Service network of weather stations throughout the country includes climate stations that have daily records of temperature and rainfall dating as far back as 1781 (DWD 2013a). Parallel to weather observations, phenological records have been collected since 1951, including the annual onset of sowing/planting for several field crops (DWD 2013b). The local nature of both weather and phenology data results in incomplete time series when observational stations are decommissioned or moved, so that of the 1965 phenological observatories with some record of maize planting dates in the years 1993–2012, the number of stations with all 20 years was 435. Schaber et al. (2010) showed how to reconcile the 'messy' phenological records, but also, because of the considerable amount of data, time series can be filtered by strict standards to remove incomplete or erroneous series, while still maintaining sufficient spatial and temporal variability in the

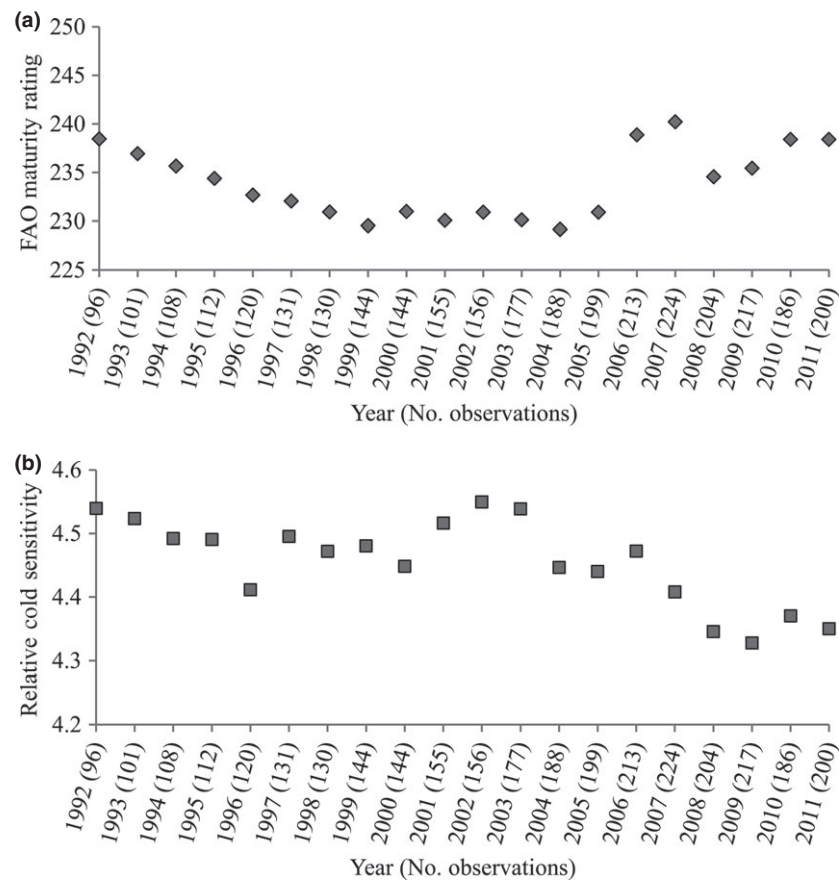


Fig. 1 Recent trends in maize breeding: (a) FAO maturity rating (relating moisture content to photothermal accumulation, roughly equivalent to days between planting and maturity) and (b) Index values for cold sensitivity (opposite of early vigour), year and number of observations along the x-axis.

Table 1 Maize planting day summary statistics by selected criteria

	All	1993–2002	2003–2012	≥250 m a.s.l.	<250 m a.s.l.	≥51°N	<51°N	≥8.5°E	<8.5°E
Mean	116.5	118.4	114.7	117.4	115.6	116.2	116.7	116.2	117.2
sta. dev.	7.80	7.18	7.97	8.29	7.12	7.08	8.23	7.3	9.13
Trend	−0.31	0.06	−0.29	−0.26	−0.34	−0.37	−0.26	−0.30	−0.32
P-value of trend	0.000	0.102	0.000	0.000	0.000	0.000	0.000	0.000	0.000

remaining dataset (see below). After looking for a causal relationship between weather and phenology, the two were used together to elucidate their influence on a third dataset, a time series of yields at district (*Kreis*) scale, (an administrative scale between municipality and county).

Maize phenology

To have a thorough geographic distribution of data, including records from the north-eastern states of the country, 1993 was selected as the start of time series data, including only stations with 20 years of uninterrupted observations of maize planting dates recorded as occurring before day 180, that is, before July. The time period 1993–2012 also coincides with the onset of a distinct climatic reference period (1991–2020) as defined by the World Meteorological Association (WMO 2013). With the above filters imposed, 432 stations containing 8640 data points for planting dates became the basis for analysis. Table 1 uses these data to show how planting occurred earlier in the more recent decade (2013–2012), at lower elevations and to the North and East. Variance increases recently, at higher elevations and to the South and West.

Figure 2 compares the distribution of planting dates between the two decades 1993–2012 and 2003–2012, including the broadening distribution and shift earlier

during the recent decade, with a noticeable gap at day 121, the first of May holiday.

Weather

For the initial model, to explain maize planting dates with spring weather, explanatory records were chosen to be limited to weather stations without gaps in daily mean temperature, minimum temperature or precipitation measurements. Although winter weather directly affects planting via spring soil conditions (Van Oort et al. 2012), it was assumed that weather during March, April and May are primarily responsible for the date of maize planting in a respective year. Based on this, weather records for the relevant growing season were compiled, so that weather data were used only from stations without missing average or minimum temperature or precipitation between days 57 (Feb. 26th) and 294 (≈ Oct. 21st) throughout the years 1993–2012. This provided 160 stations throughout the country. These data series were partitioned into 42 one-week periods, for which the minimum and mean temperatures and precipitation were established as explanatory variables. The first model, explaining planting dates with weekly weather, uses the first 20 weeks and the second model, relating district yields to planting dates and weather, uses weeks 21–42.

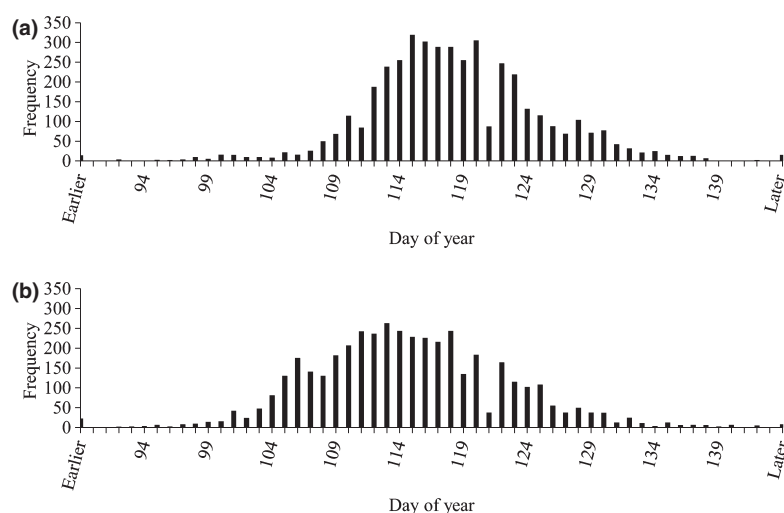


Fig. 2 Distribution of observed maize planting dates, by day of year, of a total of 4319 observations: (a) 1993–2002 and (b) 2003–2012. The significance of mean planting dates being earlier in the latter period is $P = 0.15$.

If the minimum ambient temperature for growth of maize is 10 °C, this threshold is reached on average, throughout the country in week 17, in the centre of the distribution of planting dates in Figure 2.

Phenology and weather stations were collated with ArcGIS based on Euclidian surface distance and difference in elevation. To account for the influence of elevation on temperature/phenology (Jochner et al. 2012), metres of elevation were weighted ten-to-one vs. kilometres of distance and in this way each phenology station was assigned a weather station to provide explanatory weekly precipitation and temperature data, with multiple phenology stations able to share one weather station. This and a consequent filter limiting distance and difference in elevation between phenology and weather stations (below) reduced the number of weather stations to 115.

The final phenology and weather stations are shown in Figure 3.

Weather trends

Recent climate change in Europe has entailed a general warming as well as increased temperature variability, both of which are expected to continue in the future (Zebisch et al. 2005, Fischer and Schär 2009). The general warming trend masks one component of local weather that is relevant to maize cultivation, minimum spring temperatures. Kim et al. (2014) found fewer but more extreme frost days occurring in recent decades in the conterminous US. Similarly, observations from the final 115 weather stations confirm the warming trend but also indicate decreasing minimum temperatures during four consecutive weeks

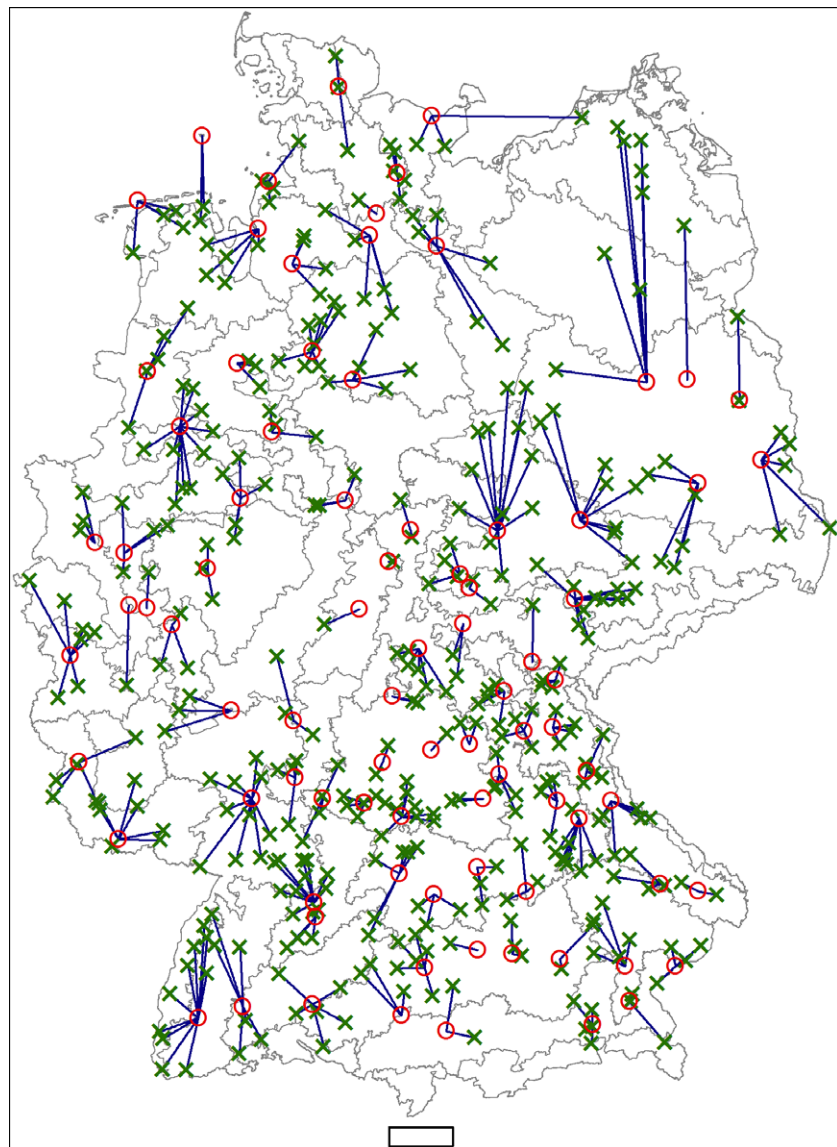


Fig. 3 Soil-climate zones (outlines), phenology (X) and weather (O) stations, with lines adjoining explanatory pairs. Bar at bottom represents 50 km. Source: Own Compilation based on Josberg et al. (2007), DWD (2013a,b). [Colour figure can be viewed at wileyonlinelibrary.com]

(17–20) at the latter half of the planting period. In most of the 16 federal states, over the two recent decades, weekly minimum temperatures during weeks 17 and 20 decreased by around 1 °C every 10 years, at a minimum of 0.05 degrees of significance. In weeks 18 and 19, the decrease was observable but not as dramatic, and the combination of a large number of observations, small magnitudes of change and low levels of significance make any conclusive statement difficult. Table A1 has coefficients, by federal state, for minimum temperature changes during the period.

While the trend towards more extreme low temperatures during these weeks is only significant for some of the federal states, it shows a consistent decrease in minimum temperatures around the typical time for planting, a clear increase in one major source of abiotic stress for maize farmers throughout the country. This poses an increased risk of fatal or permanent damage to young maize stands. This abiotic risk alone would encourage a farmer to plant maize later in the year. In conjunction with risk-mitigating actions such as cultivar choice, the more extreme cold snaps may have only limited the progression of planting dates towards earlier in the year, as the general warming trend would provoke.

District yields

The German Federal Statistics Office compiles regional (*Länder*) records of crop yields at district (*Kreise*) scale. There are a combined 475 rural and urban districts covering between 36 and 5470 km² with a mean of 888 km² (DESTATIS 2014a). Records on silage maize fresh-matter yields are available for less than half of the districts in the country throughout the examined time period. Time series with yields greater than zero and no gaps are available from 1993 to 2012 for 164 districts (DESTATIS 2014b). This increases to 253 for the period 1994–2012 so the latter period was chosen to increase both geographic range and number of observations. Two separate districts contain questionable minima of 4.74 Mg ha⁻¹ in 1996 and 8.64 Mg ha⁻¹ in 2008, so the two districts were dropped, leaving a minimum of 17 Mg h⁻¹, recorded in both 2003 and 2006, 2 years of extreme weather. The lowest mean yield for the whole country was 40.6 Mg ha⁻¹ in 2003, and the highest was 50 Mg ha⁻¹ in 2011, which also had the highest yearly maximum at 72.4 Mg ha⁻¹. Some of this positive trend in yields can be attributed to farmer adaptation to the dynamic production risk posed by abiotic stresses. The general increase in temperatures together with successful breeding for improved early vigour has permitted earlier planting of maize, facilitating a longer growing season, and also the coordination of specific crop development phases to observed weather patterns, such as to achieve flowering before early summer drought (Hund et al. 2004).

Panel regression of planting dates

It was undertaken to model maize planting dates with weekly weather and binary indicators of soil-climate zone in which phenology stations are located. First, simple regressions were run to test which restrictions of distance and elevation difference produced the most accurate results with Akaike's (1974) information criterion, divided by the number of observations: $AIC/n = -2\ln L + 2k/n$ where $\ln L$ is the maximised log-likelihood of the model, k is the number of parameters estimated and n is the number of observations. This method determined that limiting the distance between weather and phenology stations to 77.5 surface km and 80 vertical m produced the best model version. This reduced the number of phenology stations to 374 and weather stations to 86, with a total of 7480 observations, which were then partitioned into panels by phenology station and, because of the recurring binary variables, run as a random effects panel regression, defined by STATA (2011) as: $y_{it} - \theta \bar{y}_i = (1 - \theta)\alpha + (x_{it} - \theta \bar{x}_i)\beta + \{(1 - \theta)v_i + (\epsilon_{it} - \theta \bar{\epsilon}_i)\}$ where y_{it} denotes the day of planting at the i th phenology station in year t , the x_{it} are conditioning variables for the i th station in year t , v_i is a random error unique to the i th station with variance σ_v^2 , and ϵ_{it} is a random error that varies over phenology stations and time with variance σ_ϵ^2 . Temporal dependence within a phenology station is represented by the correlation coefficient $\rho = \sigma_v^2 / (\sigma_\epsilon^2 + \sigma_v^2)$ such that the larger the parameter θ , the greater this correlation. This may be seen by considering the relationship $\theta = 1 - \sigma_\epsilon / (\sigma_\epsilon^2 + T\rho(\sigma_\epsilon^2 + \sigma_v^2))^{1/2}$, where T is the number of temporal observations. Alternatively, as it is also the case that $\theta = 1 - \sigma_\epsilon / (\sigma_\epsilon^2 + T\sigma_v^2)^{1/2}$, where T is the number of temporal observations, a value of zero for θ implies that the variance of the phenology-specific random errors is zero and hence there is no temporal dependence within the cross-section. (For further explanation see for example Greene 2011).

The day of planting was thus regressed against year (after 1992), mean temperatures and precipitation during weeks 9 through 20 (as twk9-20 and pwk9-20, respectively) and all but one of the indicator variables for soil-climate zone (SCZ).

Panel regression of yields

A subset of the above phenology and associated weather records were related to district-level yields. Included districts with 19 years each produced a total of 2831, which were regressed against the mean planting date at stations within each district, along with the weekly mean and minimum temperatures and mean precipitation at stations allocated to these phenology stations, and the soil-climate zone (SCZ) encompassing each district. In a subsequent

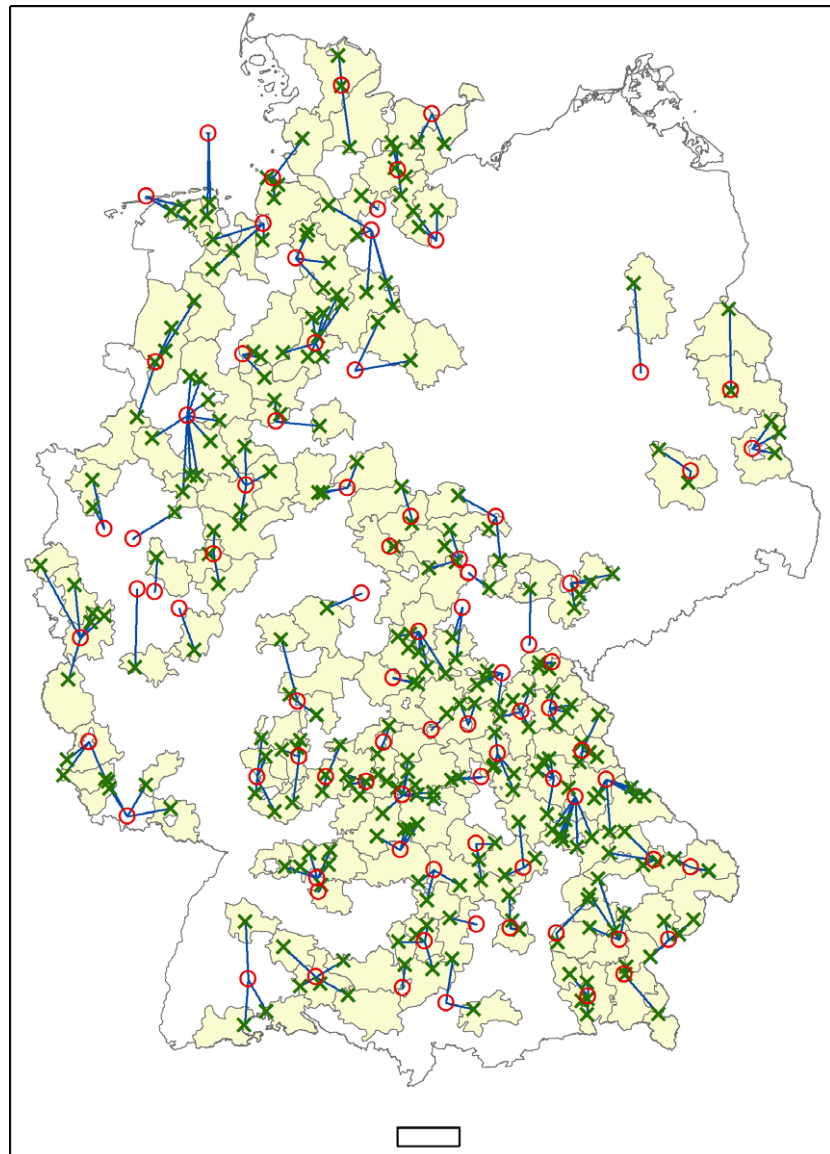


Fig. 4 Districts (outlined) and Phenology(X) and weather (O) stations used in yield model. Bar at bottom represents 50 km. Source: Own compilation based on Josberg et al. (2007), DWD (2013a,b). [Colour figure can be viewed at wileyonlinelibrary.com]

regression, deviations of planting date from district means, and its square root, were used to derive a non-linear relationship between planting date and yield. While this model is not as robust, it allows theoretical optimisation. The districts, phenology and weather stations used are in Figure 4.

The indicator for SCZ199 was again omitted from the regression, while all others were retained.

Results

Planting dates

Using the 7480 observations, the planting date regression model achieved an overall R^2 of 0.28 and a RMSE of 5.36 days. Table A2 has the complete results from this regression, showing that weekly temperatures are mostly

highly significant and have negative coefficients, indicating that **higher temperatures encourage farmers to plant earlier**. This could be a combination of more rapid drying of the soil to allow traffic and an indicator to farmers that **maize can be planted with relatively low risk of cold damage**. Precipitation during weeks 14 through 17, the middle of the normal planting period is significantly explanatory of the same year's planting date. Positive coefficients show how rain delays planting, partly due to the diminished ability of a wet soil to withstand field traffic.

District yields

Regression of the 2831 maize yield data against mean district planting date and weekly weather variables produced an R^2 of 0.62 and RMSE of 3.41 Mg ha⁻¹. An annual trend

was indicated at $0.34 \text{ Mg ha}^{-1} \text{ year}^{-1}$, but was not significant. Number of years after 1994 and its square (both highly significant) were substituted for it. Overall model accuracy was comparable before and after the substitution. This quadratic function of yield with respect to year produced modelled yields that slightly decrease from 1994 to 2002 and increasing yields thereafter. This may be tied to the increasing prevalence of earlier ripening cultivars in the first decade that reversed in the second (see Fig. 1a). The highly significant mean planting date indicated an increase of 46 kg ha^{-1} for every day earlier that the crop is planted. Table A3 has the complete results of the yield regression model. Where mean weekly temperatures are significant, coefficients alternate between being positive and negative. Mean weekly precipitation, where significant, is predominantly positive, as is the case with minimum weekly minimum temperatures, not surprisingly, as more rainfall and less extreme cold temperatures should be expected to boost yields.

Optimal planting dates

Crop growth simulation can be used to determine the optimal planting dates for maize (e.g. Saseendran et al. 2005) or a weather-based algorithm can be used (Elnesr et al. 2013). The above linear coefficient relating yield to planting date is valuable in its simplicity and should be accurate to within a few days of divergence from the norm. However, it is unreasonable to assume that the relationship continues more than a few days in either direction. In order to obtain theoretically optimal planting dates, the planting date variable was expanded into d and $d^{1/2}$, the first indicating how many days each year's district mean planting day preceded the 19-year mean, and another as its square root. Their significance was first tested individually by substituting each for mean planting date in the above regression while omitting the year variables. The coefficient and significance were, respectively, 0.051 Mg ha^{-1} and $P = 0.051$ for days prior and 0.21 Mg ha^{-1} and $P = 0.007$ for the root. Both variables were then included together in the same regression producing equivalent overall model accuracy and a function for Mg of yield, for all included districts, $y = -0.0827d + 0.4318d^{1/2} + c$ where d is days prior to the station mean and c is a constant encompassing all other variables. Yield maximised with respect to days prior is thus $D^* = (0.5 \times 0.4318/0.0827)^2$ or 6.82 days. The latter regression performed by federal state produced results ranging from 0.92 days earlier in Bavaria, to 16.91 days earlier in Baden-Württemberg. However, coefficients for d and $d^{1/2}$ were only significant for some states. This could indicate some of the technical inefficiency in the yield gap, or demonstrate how the model captures farmer behaviour better in a certain region.

Discussion

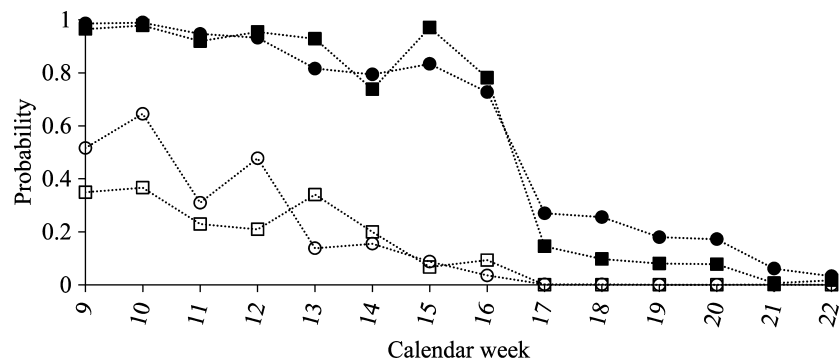
Estimating the nature of adaptive field crop management is an important theme in climate change impact studies. And while the expansive phenological dataset of the German weather service has proven to be a valuable tool in exploring the relationships between climate and vegetation (e.g. Schaber and Badeck 2005, Englert et al. 2008), less work has focused on the human influence on landscapes, as driven by weather. Crop management, especially planting dates, ultimately has a profound effect on landscape function by way of farmers seeking to increase yields and/or decrease yield uncertainty. Difficulty arises in the application of principles derived from site-specific interactions between weather, soil and management to the regional scale where geographic heterogeneity strains the plausibility of behavioural assumptions. To confidently estimate planting dates throughout a country the size of Germany, a balance must be found between geographic detail which is of limited availability, and aggregation to impose homogeneity.

Mirschel et al. (2014) with the YIELDSTAT model showed how a statistical model can offer advantages over process-based simulation models when seeking yield predictions at the regional scale. The model developed here uses less specific data, but it is available throughout Germany. Not to be compared with YIELDSTAT as a yield prediction model, the intention here was to demonstrate how farmers planting maize respond to early season weather, and how this affects eventual yields. Spring weather was shown to reliably determine planting dates that influence yields, while one could skip the determination of planting dates and use spring weather as direct drivers of yield. This would, however, omit an important mechanism by which Germany farmers may adapt to climate using the facilities, that is cultivars, at their disposal. Also, implications of planting dates such as land cover dynamics and field work requirements/availability are influenced directly by crop management independently of yield.

Germany-wide, planting 7 days earlier, according to the implied production function, would increase the per-hectare yield by 563 kg, or about one per cent of typical yield.

At a price of 30 Euros per ton, this provides approximately 17 Euros of additional revenue per hectare but implies potential variable cost increases in fertiliser, depending on the combination of organic and mineral fertilisers, (e.g. additional nitrogen at the rate of 0.1 % of marginal yield or 5.63 kg ha^{-1}) and transportation, depending on silage delivery terms (KTBL 2012). Additionally, earlier planting leads to increased risk of damaging cold weather early in the growing season. Within the 7480 weekly weather observations throughout the country (section Panel regression of planting dates), the probability of a frost (min $T \leq 0^\circ \text{C}$) increases from 0.09 % during week 17 to

Fig. 5 Empirical probability of crop-stressful cold weather, during each week of the analysis, at two temperature thresholds and over two time periods. Solid symbols: $P(\min T) \leq 5^\circ\text{C}$; Hollow: $P(\min T) \leq 0^\circ\text{C}$; Circles: 1993–2002; Squares: 2003–2012.



6.47 % in week 16, respective probabilities that temperatures drop below 5°C are 21 % and 75 %. Figure 5 shows the empirical probability of weekly minimum temperatures dropping below 5°C (upper lines) and 0°C (lower lines) by week and comparing decades.

The occurrence of lethal low temperatures may be damaging enough that replanting, at a cost of 200 Euros ha^{-1} (KTBL 2012), is worth the expense. This replanting cost can be combined with the increase probability of frost to partially quantify the risk inherent in a chosen planting date. For example, this additional cost, planting in week 16 instead of week 17, can be calculated as: $6.38\% \times 200 = 12.76$ Euro. Non-deadly but damaging low temperatures ($<5^\circ\text{C}$) can also be associated with a given level of yield loss and associated with a probability of occurrence to further assign a cost, via risk, in choosing a day to plant maize.

Observed and estimated planting dates are station-specific, but they could be used to represent the surrounding area. Yields at the district level, to some degree, are likewise applicable to neighbouring districts not included in the analysis. To obtain a statistical forecast of maize planting dates and yields in the future, simulated weather data could be used as model input, as in Schröder et al. (2014) for phenology and Ewert et al. (2005) for yields. This could provide a flexible alternative to more complicated and time-consuming methods of agent-based and crop simulation modelling such as in Aurbacher et al. (2013). Statistical models to forecast planting dates based on local weather could be the most efficient way to account for strategic management and adaptation in agroecosystem modelling, such as with process-based crop modelling where simulated yields are sensitive to planting dates (Parker et al. 2016). The same climatic data used for simulation can be quickly rendered into projections of planting dates and yields to respectively complement and accompany simulations of field crops throughout Germany, maize in this example. The use of region and location constants in the predictive model limit its applicability to Germany, an equivalent model outside the country would need a set of local

observations and regional distinctions to capture geographic characteristics.

Conclusions

A simple statistical model has been developed to predict maize planting dates and yields throughout Germany. **The models show that farmers planting maize earlier can expect greater yield, notwithstanding the increased probability of damaging cold weather.** Crop management alters the effective environmental conditions determining plant growth, and genetic advance has enabled greater managerial flexibility. While it may appear that farmers are not adequately adapting to observable climate change, here is elucidated some of the sacrifice of expected yield/revenue in favour of reduced yield/cost uncertainty. Despite generally increasing spring temperatures in Germany, evidence suggests that the risk of cold stress is not abating from the climatic side, but breeding progress is alleviating some of this risk. Some of the upward trend in silage maize yields is traceable to the fact that farmers are planting the crop earlier, in response to the likelihood of cold stress and available management options.

The methods applied above could produce respective datasets and analysis on other major German field crops. Sufficient phenological (planting date) and district yield data are available to greater or lesser extent for winter wheat, rye, barley and rapeseed, and the summer crops sugar beets, oats and potatoes. Two different sets of weekly weather data, including late summer weather for planting and the rest of the year for yields, would be applicable to all fall-planted crops, while the other summer crops, planted earlier than maize, should include earlier spring weather records to estimate planting dates.

References

- Amler, R., 2006: Produktsicherheit bei Silomais durch optimiertes Reifemanagement. *Gesunde Pflanzen* 58, 152–172.

- Aurbacher, J., P. S. Parker, G. A. Calberto Sánchez, J. Steinbach, E. Reinmuth, J. Ingwersen, and S. Dabbert, 2013: Influence of climate change on short term management of field crops – a modelling approach. *Agric. Sys.* 119, 44–57.
- Bassu, S., N. Brisson, J.-L. Durand, K. Boote, J. Lizaso, J. W. Jones, C. Rosenzweig, A. C. Ruane, M. Adam, C. Baron, B. Basso, C. Biernath, H. Boogaard, S. Conijn, M. Corbeels, D. Deryng, G. De Sanctis, S. Gayler, P. Grassini, J. Hatfield, S. Hoek, C. Izaurralde, R. Jongschaap, A. R. Kemanian, K. C. Kersebaum, S. H. Kim, N. S. Kumar, D. Makowski, C. Mueller, C. Nendel, E. Priesack, M. V. Pravia, F. Sau, I. Shcherbak, F. Tao, E. Teixeira, D. Timlin, and K. Waha, 2014: How do various maize crop models vary in their responses to climate change factors? *Glob. Change Biol.* 20, 2301–2320.
- Birch, C., J. Vos, and P. Van der Putten, 2003: Plant development and leaf area production in contrasting cultivars of maize grown in a cool temperate environment in the field. *Eur. J. Agron.* 19, 173–188.
- Bock, A., T. H. Sparks, N. Estrella, and A. Menzel, 2013: Changes in the timing of hay cutting in Germany do not keep pace with climate warming. *Glob. Change Biol.* 19, 3123–3132.
- Bonhomme, R., 2000: Bases and limits to using “degree day” units. *Eur. J. Agron.* 13, 1–10.
- BSA (Bundessortenamt-Federal Plant Variety Office), 2010, Beschreibende Sortenliste Getreide, Mais, Öl- und Faserpflanzen, Leguminosen, Rüben, Zwischenfrüchte, 2010.
- Chmielewski, F. M., A. Müller, and E. Bruns, 2004: Climate changes and trends in phenology of fruit trees and field crops in Germany, 1961–2000. *Agric. For. Meteorol.* 121, 69–78.
- Darby, H. M., and J. G. Lauer, 2002: Harvest date and hybrid influence on corn forage yield, quality, and preservation. *Agron. J.* 94, 559–566.
- Deryng, D., W. Sacks, C. Barford, and N. Ramankutty, 2011: Simulating the effects of climate and agricultural management practices on global crop yield. *Glob. Biogeochem. Cy.* 25, 1–18.
- DESTATIS Statistisches Bundesamt, 2014a. (n.d.). Retrieved May 25, 2014, from https://www-genesis.destatis.de/genesis/online/data.jsessionid=93B2B21ECC985A5973676E333643934A.tomcat_GO_2_1?operation=abrufabelleAbrufen&selectionname=11111-0002&levelindex=1&levelid=1401014421340&index=2
- DESTATIS Statistisches Bundesamt, 2014b. (n.d.). Retrieved March 15, 2014, from <https://www.regionalstatistik.de/genesis/online/data.jsessionid=F9080F0E24B1DDB3B2BDFCF43607F250?operation=abrufabelleAbrufen&selectionname=115-46-4&levelindex=1&levelid=1394914247507&index=1>
- DWD, 2013a: Weather and Climate -Deutsche Wetterdienst – Agriclimatology. German Weather Service, Offenbach, Germany. <http://www.dwd.de/> [last accessed 19 July 2016].
- DWD, 2013b: Weather and Climate – Deutsche Wetterdienst – Phenology. German Weather Service, Offenbach, Germany. http://www.dwd.de/DE/klimaumwelt/klimaueberwachung/phaenologie/daten_deutschland/daten_deutschland_node.html [last accessed 19 July 2016].
- Eitzinger, J., M. Trnka, D. Semerádová, S. Thaler, E. Svobodová, P. Hlavinka, B. Šiška, J. Takáč, L. Malatinská, M. Nováková, M. Dubrovský, and Z. Žalud, 2013: Regional climate change impacts on agricultural crop production in Central and Eastern Europe–hotspots, regional differences and common trends. *J. Agr. Sci.* 151, 787–812.
- Elnesr, M. N., A. A. Alazba, and A. A. Alasadon, 2013: An arithmetic method to determine the most suitable planting dates for vegetables. *Comput. Electron. Agr.* 90, 131–143.
- Englert, C., R. Pesch, G. Schmidt, and W. Schröder, 2008: Analysis of spatially and seasonally varying plant phenology in Germany. *Geospatial Crossroads GI_Forum* 8, 81–89.
- Estrella, N., T. H. Sparks, and A. Menzel, 2007: Trends and temperature response in the phenology of crops in Germany. *Glob. Change Biol.* 13, 1737–1747.
- Ewert, F., M. D. A. Rounsevell, I. Reginster, M. J. Metzger, and R. Leemans, 2005: Future scenarios of European agricultural land use: I. Estimating changes in crop productivity. *Agr. Ecosyst. Environ.* 107, 101–116.
- Fairey, N. A., 1983: Yield, quality and development of forage maize as influenced by dates of planting and harvesting. *Can. J. Plant Sci.* 63, 157–168.
- Farooq, M., T. Aziz, Z. A. Cheema, M. Hussain, and A. Khaliq, 2008a: Activation of antioxidant system by KCl improves the chilling tolerance in hybrid maize. *J. Agron. Crop Sci.* 194, 438–448.
- Farooq, M., T. Aziz, S. M. A. Basra, M. A. Cheema, and H. Rehman, 2008b: Chilling tolerance in hybrid maize induced by seed priming with salicylic acid. *J. Agron. Crop Sci.* 194, 161–168.
- Farooq, M., T. Aziz, M. Hussain, H. Rehman, K. Jabran, and M. B. Khan, 2008c: Glycinebetaine improves chilling tolerance in hybrid maize. *J. Agron. Crop Sci.* 194, 152–160.
- Fischer, E. M., and C. Schär, 2009: Future changes in daily summer temperature variability: driving processes and role for temperature extremes. *Clim. Dynam.* 33, 917–935.
- Greaves, J. A., 1996: Improving suboptimal temperature tolerance in maize-the search for variation. *J. Exp. Bot.* 47, 307–324.
- Greene, W. H., 2011: *Econometric Analysis*, 7th edn. Prentice-Hall, New Jersey, NJ, USA. ISBN 978-0131395381.
- Hattermann, F. F., Z. W. Kundzewicz, S. Huang, T. Vetter, F.-W. Gerstengarbe, and P. Werner, 2013: Climatological drivers of changes in flood hazard in Germany. *Acta Geophys.* 61, 463–477.
- Hund, A., Y. Fracheboud, A. Soldati, E. Frascaroli, S. Salvi, and P. Stamp, 2004: QTL controlling root and shoot traits of maize seedlings under cold stress. *Theor. Appl. Genet.* 109, 618–629.
- Jochner, S. C., T. H. Sparks, N. Estrella, and A. Menzel, 2012: The influence of altitude and urbanisation on trends and mean dates in phenology (1980–2009). *Int. J. Biometeorol.* 56, 387–394.
- Kaukoranta, T., and K. Hakala, 2008: Impact of spring warming on sowing times of cereal, potato and sugar beet in Finland. *Agr. Food Sci.* 17, 165–176.

- Kim, Y., J. Kimball, K. Didan, and G. Henebry, 2014: Response of vegetation growth and productivity to spring climate indicators in the conterminous United States derived from satellite remote sensing data fusion. *Agr. For. Meteorol.* 194, 132–143.
- Knappenberger, T., and K. Köller, 2012: Spatial assessment of the correlation of seeding depth with emergence and yield of corn. *Precis. Agric.* 13, 163–180.
- Kolo, E., F. O. Takim, and O. Fadayomi, 2012: Influence of planting date and weed management practice on weed emergence, growth, and yield of maize (*Zea mays* L.) in southern Guinea savanna of Nigeria. *J. Agr. Biodiv. Res.* 1, 33–42.
- KTBL (Kuratorium für Technik und Bauwesen in der Landwirtschaft) (ed.), 2012: *Betriebsplanung Landwirtschaft 2012/13*. KTBL-Datensammlung, Darmstadt, Germany.
- Kucharik, C. J., 2006: A multidecadal trend of earlier corn planting in the central USA. *Agron. J.* 98, 1544–1550.
- Kucharik, C. J., 2008: Contribution of planting date trends to increased maize yields in the central United States. *Agron. J.* 100, 328–336.
- Leclère, D., P.-A. Jayet, and N. de Noblet-Ducoudré, 2013: Farm-level autonomous adaptation of European agricultural supply to climate change. *Ecol. Econ.* 87, 1–14.
- Leenhardt, D., and P. Lemaire, 2002: Estimating the spatial and temporal distribution of sowing dates for regional water management. *Agri. Water Manage.* 55, 37–52.
- Ma, S., G. Churkina, and K. Trusilova, 2012: Investigating the impact of climate change on crop phenological events in Europe with a phenology model. *Int. J. Biometeorol.* 56, 749–763.
- Mast, B., W. Claupein, and S. Graeff-Hönninger, 2014: Using a crop growth model to quantify regional biogas potentials: an example of the model region Biberach (south-west Germany). *Bioenergy Res.* 7, 1–12.
- Maton, L., D. Leenhardt, and J. Bergez, 2007: Geo-referenced indicators of maize sowing and cultivar choice for better water management. *Agron. Sustain. Dev.* 27, 377–386.
- Menzel, A., T. H. Sparks, N. Estrella, E. Koch, A. Aasa, R. Ahas, K. Alm-Kühler, P. Bissolli, O. G. Braslavská, A. Briede, and F. M. Chmielewski, 2006a: European phenological response to climate change matches the warming pattern. *Glob. Change Biol.* 12, 1969–1976.
- Menzel, A., J. von Vopelius, N. Estrella, C. Schleip, and V. Dose, 2006b: Farmers' annual activities are not tracking the speed of climate change. *Clim. Res.* 32, 201.
- Mirschel, W., R. Wieland, K.-O. Wenkel, C. Nendel, and C. Guddat, 2014: YIELDSTAT – a spatial yield model for agricultural crops. *Eur. J. Agron.* 52, 33–46.
- Nendel, C., K. Kersebaum, W. Mirschel, and K. Wenkel, 2014: Testing farm management options as climate change adaptation strategies using the MONICA model. *Eur. J. Agron.* 52, 47–56.
- Olesen, J. E., C. D. Børgesen, L. Elsgaard, T. Palosuo, R. Rötter, A. O. Skjelvåg, P. Peltonen-Sainio, T. Börjesson, M. Trnka, F. Ewert, S. Siebert, N. Brisson, J. Eitzinger, E. D. van Asselt, M. Oberforster, and H. J. van der Fels-Klerx, 2012: Changes in time of sowing, flowering and maturity of cereals in Europe under climate change. *Food Addit. Contam. Part A* 29, 1527–1542.
- Oluwaranti, A., M. Fakorede, and B. Badu-Apraku, 2008: Grain yield of maize varieties of different maturity groups under marginal rainfall conditions. *J. Agric. Sci. Belgrade* 53, 183–191.
- Parker, P., J. Ingwersen, P. Högy, E. Priesack, and J. Aurbacher, 2016: Simulating regional climate-adaptive field cropping with fuzzy logic management rules and genetic advance. *J. Agric. Sci.* 154, 207–222.
- Reif, J. C., S. Hamrit, M. Heckenberger, W. Schipprack, H. P. Maurer, M. Bohn, and A. E. Melchinger, 2005: Trends in genetic diversity among European maize cultivars and their parental components during the past 50 years. *Theor. Appl. Genet.* 111, 838–845.
- Roßberg, D., V. Michel, R. Graf, and R. Neukampf, 2007: Boden-Klima-Räume und Anbauggebiete als Basis des regionalisierten Sortenwesens in Deutschland. *Mitteilungen Der Landesforschungsanstalt Für Landwirtschaft Und Fischerei Mecklenburg-Vorpommern, Gülzow* (37), 24–30.
- Sacks, W. J., and C. J. Kucharik, 2011: Crop management and phenology trends in the U.S. Corn Belt: impacts on yields, evapotranspiration and energy balance. *Agr. For. Meteorol.* 151, 882–894.
- Sacks, W. J., D. Deryng, J. A. Foley, and N. Ramankutty, 2010: Crop planting dates: an analysis of global patterns. *Glob. Ecol. Biogeogr.* 19, 607–620.
- Saseendran, S. A., L. Ma, D. Nielsen, M. Vigil, and L. Ahuja, 2005: Simulating planting date effects on corn production using RZWQM and CERES-Maize models. *Agron. J.* 97, 58–71.
- Schaber, J., and F. W. Badeck, 2005: Plant phenology in Germany over the 20th century. *Reg. Environ. Change* 5, 37–46.
- Schaber, J., F. Badeck, D. Doktor, and von Bloh W., 2010: Combining messy phenological time series. In: I.L. Hudson, and M.R. Keatley, eds. *Phenological Research*, pp. 147–158. Springer, Dordrecht, the Netherlands.
- Schröder, W., G. Schmidt, and S. Schönrock, 2014: Modelling and mapping of plant phenological stages as bio-meteorological indicators for climate change. *Environ. Sci. Eur.* 26, 1.
- Siebert, S., and F. Ewert, 2012: Spatio-temporal patterns of phenological development in Germany in relation to temperature and day length. *Agr. For. Meteorol.* 152, 44–57.
- Staggenborg, S. A., D. L. Fjell, D. L. Devlin, W. B. Gordon, L. D. Maddux, and B. H. Marsh, 1998: Selecting optimum planting dates and plant populations for dryland corn in Kansas. *J. Prod. Agr.* 12, 85–90.
- StataCorp, 2011: *Stata: Release 12*. Statistical Software. Stata-Corp LP, College Station, TX, USA.
- Stone, P. J., I. B. Sorensen, and P. D. Jamieson, 1999: Effect of soil temperature on phenology, canopy development, biomass and yield of maize in a cool-temperate climate. *Field Crops Res.* 63, 169–178.

- Strigens, A., C. Grieder, B. I. G. Haussmann, and A. E. Melchinger, 2012: Genetic variation among inbred lines and test-crosses of maize for early growth parameters and their relationship to final dry matter yield. *Crop Sci.* 52, 1084–1092.
- Troost, C., and T. Berger, 2014: Dealing with uncertainty in agent-based simulation: farm-level modeling of adaptation to climate change in Southwest Germany. *Am. J. Agric. Econ.* 97, 833–854.
- Van Oort, P., B. Timmermans, and A. van Swaaij, 2012: Why farmers' sowing dates hardly change when temperature rises. *Eur. J. Agron.* 40, 102–111.
- Waha, K., L. Van Bussel, C. Müller, and A. Bondeau, 2012: Climate-driven simulation of global crop sowing dates. *Glob. Ecol. Biogeogr.* 21, 247–259.
- White, J. W., G. Hoogenboom, B. A. Kimball, and G. W. Wall, 2011: Methodologies for simulating impacts of climate change on crop production. *Field Crops Res.* 124, 357–368.
- WMO, World Meteorology Organization, 2013: Statement on the status of the global climate in 2012. WMO-No. 1108 ISBN: 978-92-63-11108-1. p. 13.
- Zebisch, M., T. Grothmann, D. Schröter, C. Hasse, U. Fritsch, and W. Cramer, 2005: Climate change in Germany. Vulnerability and adaptation of climate sensitive sectors/ Klimawandel in Deutschland-Vulnerabilität Und Anpassungsstrategien Klimasensitiver Systeme. Federal Environmental Agency Germany/Umweltbundesamt, Dessau, Report, 201, 41–253.

Appendix A

Table A1 Trends in weekly minimum temperatures in degrees Celcius (°C) per year (yr) by Federal State 1993–2012, *t*-test P-values ($P>|t|$) of the regression coefficient of the linear trend, and number of observations (Obs.), a product of station-count in a state and the 20 years

State	Obs.	Week 17		Week 18		Week 19		Week 20	
		°C yr ⁻¹	$P> t $	°C yr ⁻¹	$P> t $	°C yr ⁻¹	$P> t $	°C yr ⁻¹	$P> t $
Brandenburg	140	−0.134	0.00	−0.027	0.45	−0.058	0.09	−0.141	0.00
Baden-Württemberg	360	−0.116	0.00	−0.104	0.00	−0.038	0.10	−0.130	0.00
Bavaria	700	−0.146	0.00	−0.073	0.00	−0.031	0.05	−0.216	0.00
Bremen	40	−0.081	0.13	0.012	0.82	−0.030	0.66	−0.058	0.36
Hessen	140	−0.118	0.00	−0.070	0.04	−0.051	0.15	−0.139	0.00
Hamburg	40	−0.136	0.01	−0.034	0.59	−0.075	0.21	−0.101	0.07
Mecklenburg-Vorpommern	40	−0.178	0.00	−0.075	0.27	−0.055	0.39	−0.139	0.02
Lower Saxony	160	−0.102	0.00	−0.017	0.58	−0.048	0.13	−0.079	0.01
North Rhine-Westphalia	200	−0.101	0.00	−0.039	0.14	−0.067	0.03	−0.117	0.00
Rhineland-Palatinate	60	−0.086	0.03	−0.057	0.17	−0.036	0.54	−0.128	0.01
Schleswig-Holstein	80	−0.071	0.02	−0.022	0.58	−0.049	0.23	−0.066	0.07
Saarland	20	−0.033	0.60	−0.082	0.25	0.000	1.00	−0.119	0.19
Saxony	100	−0.154	0.00	−0.027	0.46	−0.036	0.36	−0.142	0.00
Saxony-Anhalt	60	−0.150	0.00	−0.041	0.45	−0.029	0.55	−0.114	0.02
Thuringia	160	−0.130	0.00	−0.071	0.02	−0.018	0.54	−0.117	0.00

Source: DWD (2014a)

Table A2 Coefficients, standard errors and levels of significance for planting dates regressed against year, mean weekly temperatures (t) and precipitation (p) and soil-climate zone (SCZ)

Variable	Coef.	Std. err.	P> z	Variable	Coef.	Std. err.	P> z
Constant	118.163	3.260	0.000	SCZ113	7.349	3.142	0.019
year	−0.187	0.019	0.000	SCZ114	7.040	3.109	0.024
twk9	−0.146	0.029	0.000	SCZ115	4.456	3.190	0.162
twk10	−0.192	0.041	0.000	SCZ116	3.215	3.258	0.324
twk11	−0.178	0.048	0.000	SCZ117	6.577	3.371	0.051
twk12	−0.101	0.044	0.022	SCZ120	6.456	4.267	0.130
twk13	0.017	0.036	0.629	SCZ121	6.805	3.118	0.029
twk14	−0.174	0.038	0.000	SCZ122	7.794	3.376	0.021
twk15	−0.233	0.055	0.000	SCZ123	4.297	3.336	0.198
twk16	−0.513	0.048	0.000	SCZ127	12.681	3.306	0.000
twk17	−0.146	0.061	0.016	SCZ128	9.612	3.375	0.004
twk18	−0.013	0.046	0.785	SCZ129	7.394	3.900	0.058
twk19	0.140	0.038	0.000	SCZ130	8.000	3.895	0.040
twk20	−0.010	0.042	0.805	SCZ132	15.321	5.225	0.003
pwk9	0.087	0.045	0.054	SCZ133	10.918	4.267	0.011
pwk10	0.056	0.052	0.284	SCZ134	11.249	3.572	0.002
pwk11	0.071	0.043	0.095	SCZ141	3.894	3.696	0.292
pwk12	0.057	0.039	0.141	SCZ142	5.993	3.338	0.073
pwk13	0.063	0.050	0.208	SCZ143	10.877	3.423	0.001
pwk14	0.184	0.055	0.001	SCZ145	10.017	3.697	0.007
pwk15	0.309	0.044	0.000	SCZ146	8.567	3.183	0.007
pwk16	0.445	0.047	0.000	SCZ147	8.075	3.423	0.018
pwk17	0.362	0.053	0.000	SCZ148	6.854	3.375	0.042
pwk18	0.042	0.044	0.338	SCZ150	11.252	3.490	0.001
pwk19	0.018	0.032	0.578	SCZ151	7.729	3.574	0.031
pwk20	0.055	0.041	0.184	SCZ152	11.786	3.382	0.000
SCZ101	7.266	3.490	0.037	SCZ153	5.951	3.342	0.075
SCZ102	8.221	3.898	0.035	SCZ154	6.957	3.701	0.060
SCZ104	7.530	3.229	0.020	SCZ156	3.111	5.231	0.552
SCZ106	9.992	5.226	0.056	SCZ157	6.061	5.231	0.247
SCZ107	4.448	3.244	0.170	SCZ158	8.946	5.229	0.087
SCZ108	6.131	3.184	0.054	SCZ192	9.122	5.226	0.081
SCZ109	10.615	3.898	0.006	SCZ193	8.711	5.227	0.096
SCZ111	7.280	3.183	0.022	SCZ196	10.341	3.698	0.005
SCZ112	7.658	3.185	0.016	SCZ198	6.884	3.569	0.054

Table A3 Coefficients, standard errors and levels of significance for district yields, in Mg ha⁻¹, regressed against mean district planting date (day_year), year (after 1994), its square (ysq), weekly temperature (avgt) and weekly precipitation (avgp); the minimum weekly minimum temperature (minmin) of each district and the soil-climate zone (SCZ) in which it is located

Variable	Coef.	Std. err.	P> z	Variable	Coef.	Std. err.	P> z
Constant	52.173	4.801	0.000	minminwk27	0.339	0.129	0.009
day_year	-0.046	0.015	0.002	minminwk28	0.118	0.139	0.395
year	-0.367	0.143	0.010	minminwk29	0.095	0.117	0.417
ysq	0.022	0.007	0.003	minminwk30	0.068	0.127	0.591
avgtwk21	0.276	0.084	0.001	minminwk31	-0.006	0.118	0.960
avgtwk22	0.081	0.102	0.428	minminwk32	0.367	0.105	0.000
avgtwk23	0.124	0.109	0.253	minminwk33	-0.023	0.134	0.863
avgtwk24	0.227	0.122	0.064	minminwk34	-0.169	0.123	0.167
avgtwk25	-0.197	0.112	0.078	minminwk35	0.084	0.129	0.516
avgtwk26	-0.090	0.138	0.512	minminwk36	0.071	0.102	0.488
avgtwk27	-0.006	0.136	0.965	minminwk37	0.203	0.107	0.057
avgtwk28	-0.557	0.119	0.000	minminwk38	0.172	0.101	0.089
avgtwk29	0.221	0.140	0.115	minminwk39	0.120	0.119	0.313
avgtwk30	-0.433	0.132	0.001	minminwk40	0.032	0.118	0.787
avgtwk31	-0.124	0.129	0.338	minminwk41	-0.263	0.115	0.022
avgtwk32	-0.446	0.119	0.000	minminwk42	-0.048	0.123	0.696
avgtwk33	0.206	0.143	0.151	SCZ_102	-16.485	3.528	0.000
avgtwk34	0.117	0.130	0.367	SCZ_104	-16.306	2.887	0.000
avgtwk35	0.090	0.156	0.566	SCZ_106	-12.832	3.519	0.000
avgtwk36	0.465	0.122	0.000	SCZ_107	-3.044	2.802	0.277
avgtwk37	-0.296	0.127	0.020	SCZ_108	-1.473	3.065	0.631
avgtwk38	-0.589	0.142	0.000	SCZ_111	-3.954	2.671	0.139
avgtwk39	-0.165	0.135	0.222	SCZ_112	0.429	2.670	0.872
avgtwk40	0.209	0.147	0.154	SCZ_113	-0.078	2.603	0.976
avgtwk41	0.408	0.153	0.008	SCZ_114	1.326	2.579	0.607
avgtwk42	-0.039	0.161	0.809	SCZ_115	1.469	2.609	0.573
avgpwk21	-0.030	0.047	0.523	SCZ_116	1.896	2.761	0.492
avgpwk22	-0.065	0.043	0.126	SCZ_117	1.387	2.664	0.603
avgpwk23	-0.025	0.038	0.518	SCZ_120	-1.051	3.533	0.766
avgpwk24	0.063	0.045	0.160	SCZ_121	-0.232	2.692	0.931
avgpwk25	0.047	0.045	0.296	SCZ_122	-2.464	2.880	0.392
avgpwk26	-0.022	0.043	0.609	SCZ_123	-2.546	3.036	0.402
avgpwk27	0.201	0.036	0.000	SCZ_127	-2.708	2.749	0.324
avgpwk28	0.069	0.040	0.084	SCZ_128	-1.729	3.066	0.573
avgpwk29	0.078	0.035	0.025	SCZ_129	-2.299	3.580	0.521
avgpwk30	-0.065	0.043	0.132	SCZ_130	0.922	3.507	0.793
avgpwk31	0.126	0.044	0.004	SCZ_132	4.219	3.527	0.232
avgpwk32	0.016	0.032	0.620	SCZ_133	2.518	3.057	0.410
avgpwk33	0.095	0.047	0.041	SCZ_134	-6.177	2.807	0.028
avgpwk34	-0.053	0.035	0.128	SCZ_141	-1.202	2.877	0.676
avgpwk35	-0.045	0.040	0.263	SCZ_142	0.767	2.792	0.784
avgpwk36	0.043	0.046	0.347	SCZ_143	-1.658	2.891	0.566
avgpwk37	-0.026	0.041	0.527	SCZ_145	1.050	2.903	0.718
avgpwk38	-0.009	0.051	0.857	SCZ_146	-2.662	2.674	0.319
avgpwk39	0.066	0.049	0.179	SCZ_147	-1.416	3.064	0.644
avgpwk40	0.221	0.043	0.000	SCZ_148	-3.876	2.806	0.167
avgpwk41	0.091	0.062	0.143	SCZ_150	-6.127	2.884	0.034
avgpwk42	-0.153	0.073	0.036	SCZ_151	-4.314	3.571	0.227
minminwk21	0.006	0.015	0.688	SCZ_152	-4.269	3.073	0.165
minminwk22	-0.031	0.111	0.781	SCZ_153	-12.478	2.777	0.000
minminwk23	-0.094	0.116	0.417	SCZ_154	-12.200	2.922	0.000
minminwk24	-0.280	0.113	0.013	SCZ_156	-13.202	3.558	0.000
minminwk25	0.313	0.120	0.009	SCZ_192	-3.093	3.534	0.381
minminwk26	-0.131	0.113	0.246	SCZ_198	-0.523	3.043	0.864